AP Precalculus — **Unit 4, Lesson 4.2**

Group Activity — Answer Key

Name: ____________________________ Partner: _______________________ Period: _________

Remote-Control Car

A remote-control car moves along a surface. Its position in meters at time t seconds is given by the parametric function

$$f(t) = \big(x(t),\ y(t)\big) = \big(t^2 - 6t + 5,\ -t^2 + 4t\big), \quad 0 \le t \le 6,$$

where $x(t)$ measures meters east and $y(t)$ measures meters north.

Part 1 — Building the Table

1. Complete the table.

t (sec)	$x(t) = t^2 - 6t + 5$	$y(t) = -t^2 + 4t$	Position $(x,\ y)$
0	5	0	$(5,\ 0)$
1	0	3	$(0,\ 3)$
2	-3	4	$(-3,\ 4)$
3	-4	3	$(-4,\ 3)$
4	-3	0	$(-3,\ 0)$
5	0	-5	$(0,\ -5)$
6	5	-12	$(5,\ -12)$

Teacher Note

Circulate during Part 1 to ensure students are substituting into *both* components before moving on. Common error: substituting t into $x(t)$ only and leaving $y(t)$ blank.

Part 2 — Analyzing the Car's Motion

2. Leftmost east-position:
$x(t) = t^2 - 6t + 5$**; vertex at** $t = -(-6)/(2 \cdot 1) = 3$**.**
$x(3) = 9 - 18 + 5 = -4$ **m. The leftmost position is** $x = -4$ **at** $t = 3$ **s.**

3. Rightmost east-position:
Check endpoints: $x(0) = 5$ **and** $x(6) = 36 - 36 + 5 = 5$**. The rightmost position is** $x = 5$ **m (occurring at** $t = 0$ **and** $t = 6$**).**

4. Highest north-position:
$y(t) = -t^2 + 4t$**; vertex at** $t = -4/(2 \cdot (-1)) = 2$**.**
$y(2) = -4 + 8 = 4$ **m. The highest position is** $y = 4$ **at** $t = 2$ **s.**

5. When does the car cross the x-axis ($y = 0$)?
$-t^2 + 4t = 0 \ \Rightarrow\ -t(t-4) = 0 \ \Rightarrow\ t = 0$ **or** $t = 4$**.**
At $t = 0$**:** $(5, 0)$**. At** $t = 4$**:** $(-3, 0)$**.**

Teacher Note

Emphasize: zeros of $y(t)$ give x-intercepts of the curve. The coordinates are found by evaluating $x(t)$ at the t-values, not by setting $x(t) = 0$.

6. When does the car cross the y-axis ($x = 0$)?

$t^2 - 6t + 5 = 0 \Rightarrow (t-1)(t-5) = 0 \Rightarrow t = 1$ **or** $t = 5$.

At $t = 1$**:** $(0, 3)$**. At** $t = 5$**:** $(0, -5)$**.**

Teacher Note

Reinforce: zeros of $x(t)$ give y-intercepts of the parametric curve. Post the rule on the board throughout this section.

Part 3 — Extension

7. $g(t) = (\cos t, \sin t)$ for $0 \le t \le 2\pi$.

(a) Horizontal extrema:

$\cos t$ **has maximum** 1 **(at** $t = 0, 2\pi$**) and minimum** -1 **(at** $t = \pi$**). The curve extends from** $x = -1$ **to** $x = 1$**.**

(b) Vertical extrema:

$\sin t$ **has maximum** 1 **(at** $t = \pi/2$**) and minimum** -1 **(at** $t = 3\pi/2$**). The curve extends from** $y = -1$ **to** $y = 1$**.**

(c) y-intercepts (where $x(t) = \cos t = 0$):

$\cos t = 0$ **at** $t = \pi/2$ **and** $t = 3\pi/2$**.**

$y(\pi/2) = \sin(\pi/2) = 1$ **and** $y(3\pi/2) = \sin(3\pi/2) = -1$**.**

y**-intercepts:** $(0, 1)$ **and** $(0, -1)$**.**

Teacher Note

This extension introduces the unit circle as a parametric curve. Students who reach this point benefit from seeing that the same analytical tools (find zeros of a component to get intercepts) apply to non-polynomial parametric functions. No graphing is required; the answers follow directly from knowledge of the sine and cosine functions.